

Synopsis: BTFR Zero Point from Cosmological Horizon Memory

What the theorems say (without the proofs)

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

Short guide to the zero-point paper: galactic modelling already gives the BTFR *exponent*; cosmological-horizon Hysterical memory fixes $a_0 = H_0 c / 2\pi$. Proofs live in the full paper.

1 Why a zero point is needed

Galactic modelling produces $v_f^4 = K M_b$ on a capacity attractor, with $K = C^2 A^2 / y_*$. Writing $v_f^4 = G a_0 M_b$ names $a_0 = K/G$ but does not compute it. This paper supplies that absolute scale from residual-gravity memory cut off by the cosmological horizon.

2 Horizon memory clock

Euclidean regularity at a Killing horizon gives period $\beta = 2\pi/\kappa$. For the cosmological horizon, $\kappa = H_0$.

Definition 2.1 (IR memory acceleration).

$$a_{\text{IR}} := \frac{c}{\beta} = \frac{c\kappa}{2\pi}.$$

Theorem 2.2 (Cosmological memory scale).

$$a_{\text{IR}} = \frac{H_0 c}{2\pi}.$$

Raw Hubble time would have given cH_0 ; the 2π is horizon KMS regularity.

3 Subject independence

Proposition 3.1. a_{IR} is a property of the residual channel and the FLRW congruence. It does not depend on any particular probe's age or a passive basis rename.

4 Pinning and matching

Far-halo flat curves need a tiny loss rate; the softest allowed gravitational gap in expansion is the cosmological horizon memory rate (physical assumption). On the mature attractor one identifies $a_0 = a_{\text{IR}}$ (physical assumption).

Theorem 4.1 (BTFR zero point).

$$\boxed{a_0 = \frac{H_0 c}{2\pi}}, \quad v_{\text{f}}^4 = G \frac{H_0 c}{2\pi} M_{\text{b}}.$$

Corollary 4.2 (Numerics). *At Planck-like H_0 , this is $\sim 10^{-10} \text{ m s}^{-2}$, compatible with the empirical BTFR calibration.*

5 Not MOND

An effective a_0 appears, but the dynamics remain residual ordinary gravity with a ballistic halo—not a MOND interpolating function.

6 What is theorem vs assumption

- Inherited: BTFR attractor structure from galactic modelling.
- Literature: $\beta = 2\pi/\kappa$, cosmological $\kappa = H_0$.
- Theorem: $a_{\text{IR}} = H_0 c/2\pi$; final $a_0 = H_0 c/2\pi$ given the pins.
- Physical inputs: far-halo IR pin; $a_0 = a_{\text{IR}}$.
- Not claimed: MOND force law; separate microscopic C , A .